

Alwin Tay Jing Xue

Machine Learning Engineer (LLM)

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Portfolio : www.alwint.dev
GitHub : github.com/hmmm2x

PROFESSIONAL SUMMARY

Machine learning engineer with applied experience across LLM fine-tuning, retrieval systems, distributed data pipelines, and multi-tenant backend infrastructure. Built and measured a QLoRA fine-tuning pipeline, a 2.3 million row PySpark feature pipeline, and a security-hardened agentic document platform.

SKILLS

- **Programming Languages:** Python, R, Java, C++, SQL, TypeScript, JavaScript
- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn, XGBoost, QLoRA, Low-Rank Adaptation (LoRA), Parameter Efficient Fine-Tuning (PEFT), LightGBM, EfficientNet, DeiT, YOLOv8
- **LLM and Retrieval:** Retrieval Augmented Generation (RAG), hybrid search, reranking, embeddings, Qdrant, pgvector, LangChain, prompt engineering, vLLM, Ollama
- **Data and Big Data:** SQL, PySpark, Apache Spark, Extract Transform Load (ETL) pipelines, pandas, NumPy, time series analysis
- **Backend and Infrastructure:** FastAPI, PostgreSQL, Redis, row-level security, Docker, Amazon Web Services (AWS), GitHub Actions, Next.js, TypeScript
- **Spoken Languages:** English, Chinese (Mandarin), Bahasa Melayu (Malay)

PROFESSIONAL EXPERIENCE

LLM Research Assistant (Contract: 1 Year)

December 2025 - Present

Swinburne University of Technology (Sarawak Campus), Sarawak, Malaysia

- Trained a QLoRA adapter at rank 16 with 40.4 million trainable parameters, 0.92 percent of Qwen2.5 7B-Instruct, over 1,860 curated instruction examples on a single 16 GB RTX 5060 Ti at effective batch 32 and sequence length 1,536.
- Reduced the training footprint to a measured 10.4 GiB peak against roughly 112 GiB for full-parameter AdamW, through 4-bit NF4 quantization with rank-16 LoRA adapters.
- Evaluated the tutor across 9 runs on its production serving path, reaching bug-localisation accuracy of 0.813 with a 95 percent confidence interval of 0.774 to 0.846, routing accuracy of 0.941, and a debug-mode recall of 0.902.
- Engineered a streaming Python FastAPI backend with a Next.js and TypeScript client over PostgreSQL for a document reasoning application.

LLM Research Intern

June 2025 - December 2025

Swinburne University of Technology (Sarawak Campus), Sarawak, Malaysia

- Implemented a Low-Rank Adaptation bias-detection workflow orchestrated in LangChain with Ollama inference, evaluating Llama 3 8B against Direct Preference Optimization (DPO) baselines with Qdrant vector retrieval.
- Authored a research paper on a recursive refinement and bias analysis framework for conversational AI under academic supervision.

Software Business Analyst (Full-time)

Nov 2023 - June 2024

Finology Sdn Bhd, Kuala Lumpur, Malaysia

- **Analyse business requirements** and **translate them into 40+ technical Jira specifications**, defining edge cases and data validation rules to reduce developer rework by 30% across **two fintech projects (Aeon Bank & AHAM Asset Management)**.
- **Executed technical quality assurance (SIT/UAT) across staging and production environments**, isolating functional bugs in loan application workflows before deployment.
- Collaborated in an **Agile Scrum team** throughout the **SDLC**, owning **Functional Specification Document (FSD)** updates for client sign-off and applying knowledge of the software architecture to align specifications with system design..

AI/Machine Learning Intern

Oct 2023 - Nov 2023

Hexabyn Technologies Sdn. Bhd, Sarawak, Malaysia

- Develop a **Fruit Defect Detection System** using **EfficientNet-B0**, achieving **93.4% classification accuracy**.

- Curated the annotated image dataset and implemented an iterative **PyTorch training pipeline** to accelerate **model optimization**.
- Developed a **Django** interface to integrate **YOLOv8** output into a functional **quality-inspection workflow**.

Quality Control Engineer Intern (IT Programmer)

Mar 2023 - Jul 2023

LONGi Malaysia Sdn. Bhd, Sarawak, Malaysia

- Engineered a **VB.NET and MySQL automation system** for solar cell certification, processing high-volume logic to **reduce manual generation by 85% and boost engineering productivity by 60%**.
- Developed automated **Python ETL pipelines for 22 real-time datasets** and conducted **time series analysis**, deriving **data-driven insights to predict and mitigate manufacturing defects**.

TECHNICAL PROJECTS

VoidCode AI — ML Interview Preparation Platform

2025 - Present

- Built a PySpark feature pipeline processing 2.3 million code-submission rows at roughly 9,200 rows per second on 10 cores into per-learner mastery vectors for 117,000 learners.
- Trained a LambdaMART ranker over those features, reaching Recall at 100 of 0.616 against a 0.190 popularity baseline and an NDCG at 10 of 0.214.
- Authored and validated a 200-item catalogue with 125 executable problems and 699 test cases, every reference solution executing green in continuous integration.

SYNTHIEN AI — Multi-Tenant Document Intelligence Platform

2025 - Present

- Built an agentic retrieval backend across 197 source files and 79 migrations with 9 registered capabilities, hybrid search and reranking with per-claim citations, on FastAPI with PostgreSQL, Qdrant and Redis.
- Enforced fail-closed row-level security across 56 policies at the data layer, verified by an 85-test adversarial suite covering cross-tenant reads, filter injection, prompt injection and redaction before model egress.
- Instrumented PII redaction before every outbound model call and an append-only audit log spanning 88 action types, with continuous integration running lint, type-check, contract-drift and a dedicated tenancy job on Linux service containers.

EDUCATION

Swinburne University Of Technology

September 2024 - July 2026

Bachelor of Computer Science (Major in Data Science)

LICENSE & CERTIFICATION

- PyTorch for Deep Learning Professional Certificate by DeepLearning.AI [\[link\]](#)
- TensorFlow Developer Professional Certificate by DeepLearning.AI [\[link\]](#)
- AI Engineer for Developers Associate Certificate by DataCamp [\[link\]](#)
- AI Engineer for Data Scientists Associate Certificate by DataCamp [\[link\]](#)
- Data Scientist Associate Certificate by DataCamp [\[link\]](#)
- Product Management Certification by General Assembly [\[link\]](#)
- Statistical Analysis System (SAS) Certificate [\[link\]](#)

AWARDS & HONORS

- **[National]** TechFrontier Explorer × SCENT Demo Day Competition – 2nd Place Award [\[link\]](#)
- **[National]** TechFrontier Explorer × SCENT Programme – Grants Award [\[link\]](#)

REFERENCES

Available on request.